

**Detecting Illicit Cryptocurrency Transactions Using Machine Learning:
An Analytical Study Using the Elliptic Bitcoin Dataset**

Alejandro SOBA

ST3189 Machine Learning

University of London

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Abstract

This project examines whether machine learning methods can support the detection and prioritisation of potentially illicit cryptocurrency transactions using the Elliptic Bitcoin Dataset. A single real-world dataset is used to complete the three tasks required in the coursework: unsupervised learning, regression, and classification. First, Principal Component Analysis and K-means clustering are applied to explore the structure of the transaction feature space. Second, several regression models are compared using a constructed proxy risk score designed to represent relative transaction review priority. Third, binary classification models are evaluated to distinguish illicit from licit transactions among the labelled observations. The results show that the data contain meaningful structure, although unsupervised groupings do not map neatly onto transaction labels. In the regression task, Random Forest Regression outperforms linear, ridge, and lasso models. In the classification task, Random Forest Classification provides the strongest overall results, achieving high illicit recall together with very low false-positive rates. Overall, the findings suggest that machine learning can provide useful analytical support for screening and prioritising cryptocurrency transactions in an anti-money-laundering context, while also highlighting the importance of cautious interpretation.

Keywords:

machine learning, cryptocurrency, Elliptic Bitcoin Dataset, anti-money laundering, classification, clustering

1. Introduction

The growth of cryptocurrency has created new opportunities for rapid and pseudonymous financial transfer, but it has also raised concerns about the use of blockchain-based transactions in illicit activity. For institutions involved in anti-money-laundering work, financial intelligence, or compliance screening, the ability to identify transactions that deserve closer review has become increasingly important. Machine learning is relevant in this setting because it can help detect patterns that are difficult to identify through manual inspection alone, particularly in fraud detection and anti-money-laundering contexts where large transaction datasets must be screened efficiently (Baesens, Van Vlasselaer and Verbeke, 2015; Weber et al., 2019).

This coursework investigates whether machine learning methods can support the analysis of potentially illicit Bitcoin transactions using the Elliptic Bitcoin Dataset, a dataset widely used in research on cryptocurrency transaction analysis and financial crime detection (Elliptic, n.d.; Weber et al., 2019). The project is organised around the three tasks required in the module brief: unsupervised learning, regression, and classification. Rather than treating these as separate exercises, the report uses them as complementary ways of examining the same dataset. The unsupervised analysis explores whether the feature space contains meaningful structure, the regression analysis tests whether a continuous proxy risk score can be predicted, and the classification analysis evaluates the ability of different models to distinguish illicit from licit transactions. The coursework brief also places importance on clear presentation and interpretation for a reader with quantitative background but no specialist machine learning knowledge, and this report follows that approach throughout.

The analysis is guided by four research questions. First, can unsupervised learning reveal meaningful grouping patterns in the transaction data? Second, can a continuous proxy risk score be predicted with acceptable accuracy using regression methods? Third, which classification model performs best when identifying illicit transactions under class imbalance? Finally, what do the strongest-performing models suggest about the transaction features that contribute most to predictive performance?

2. Dataset and Analytical Design

The analysis uses the Elliptic Bitcoin Dataset, a real-world transaction-level dataset containing engineered numerical features and transaction class labels, which has become a common reference point in machine learning work on illicit cryptocurrency transaction detection (Elliptic, n.d.; Weber et al., 2019). After importing the feature file and joining it with the class labels, each observation represents a Bitcoin transaction described by a large set of numerical attributes and a class indicator. In the dataset, class 1 is treated as illicit, class 2 as licit, and unknown transactions are retained where relevant but excluded from supervised classification.

This structure makes the dataset well suited to the coursework brief because it allows all three required tasks to be completed within a single analytical framework. Unsupervised learning can be applied to the full feature space without relying on labels. Regression can be implemented by constructing a continuous proxy score that reflects relative transaction risk. Classification can then

focus on the labelled subset, where illicit and licit cases can be compared directly. Using one dataset throughout also improves coherence across the report and makes it easier to compare how different machine learning techniques behave when applied to the same underlying transaction environment.

Table 1 - Dataset summary

Metric	Value
Number of observations after merge	203769
Number of variables after merge	168
Labelled transactions	46564
Unknown transactions	157205
Illicit transactions (class 1)	4545
Licit transactions (class 2)	42019

A key feature of the dataset is class imbalance, which is important because minority-class problems can make accuracy alone a misleading measure of model performance (James et al., 2014). Illicit transactions form a minority of labelled observations, while many transactions are marked as unknown. This matters because high overall accuracy may still conceal weak performance on the illicit class. For that reason, later sections place particular emphasis on recall and precision rather than treating accuracy alone as decisive, since these metrics are more informative in imbalanced classification settings (James et al., 2014).

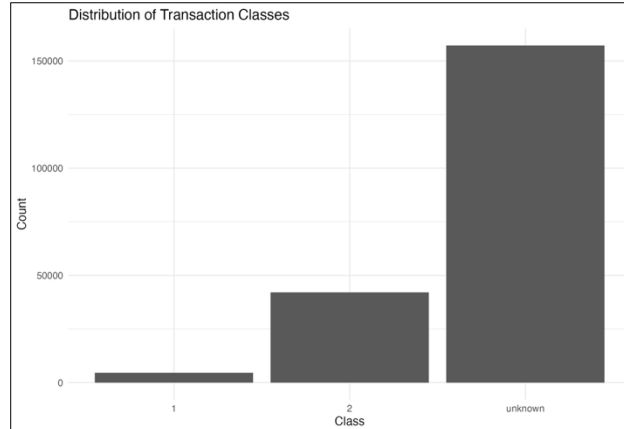


Figure 1- Distribution of transaction classes

The analytical design follows the three-task structure of the coursework. In the unsupervised stage, Principal Component Analysis is used to reduce dimensionality and K-means clustering is used to explore similarity groupings. In the regression stage, the target variable is not an observed financial outcome but a constructed proxy risk score, intended to represent review priority rather than ground-truth risk. In the classification stage, logistic regression, Support Vector Machine, and Random Forest models are compared on their ability to identify illicit transactions.

Table 2 - Methods overview

Task	Techniques	Purpose
Unsupervised Learning	PCA, K-means clustering	Explore structure and similarity groupings in the transaction feature space
Regression	Linear Regression, Ridge, Lasso, Random Forest Regression	Predict a proxy transaction risk score for analytical prioritisation
Classification	Logistic Regression, Support Vector Machine, Random Forest	Classify licit and illicit transactions

3. Unsupervised Learning

The first stage of the analysis examines whether the transaction feature space contains visible structure before any supervised modelling is applied. To do this, numeric variables were standardised and Principal Component Analysis was used to summarise the dominant directions of variation, following its standard role as a dimensionality-reduction method for high-dimensional datasets (Bishop, 2006; James et al., 2014). PCA is useful here because the feature space is high-dimensional, and a lower-dimensional representation makes it easier to inspect broad patterns without assuming a classification target, while still preserving as much variation as possible in the transformed components (Bishop, 2006; James et al., 2014).

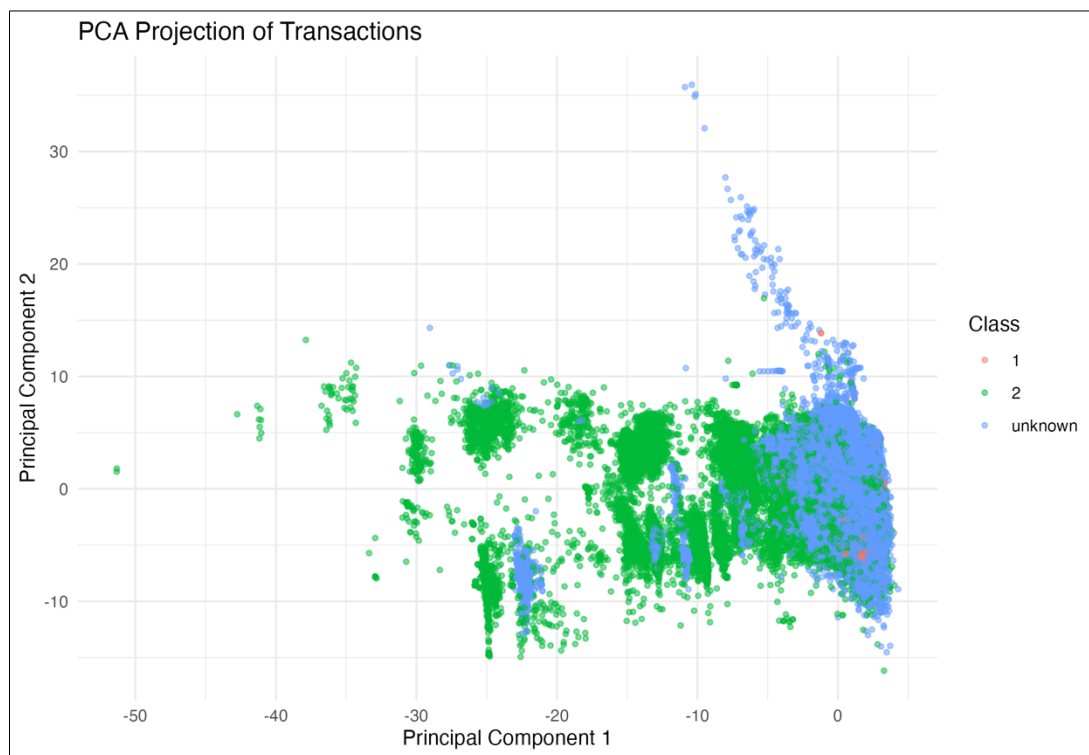


Figure 2 - PCA projection of transactions

The PCA projection suggests that the observations are not randomly scattered, and some broad structure is visible. At the same time, the known classes do not separate cleanly in the two-dimensional projection. This is not surprising. PCA preserves major variance directions in the data rather than class boundaries, so it should not be expected to produce a clean separation between illicit and licit activity in a two-dimensional projection (Bishop, 2006). The result therefore

suggests that the transaction feature space contains structure, but not one that can be reduced to a simple visual separation of criminal and non-criminal transactions.

The next step was to determine a suitable number of clusters for K-means, an unsupervised method that groups observations according to similarity in the feature space rather than known labels (James et al., 2014; Murphy, 2012). The elbow method suggested that most of the reduction in within-cluster variation occurred at relatively low values of k , with diminishing returns after approximately $k = 2$ or $k = 3$. However, the silhouette method provided a clearer decision rule, reaching its maximum at $k = 2$. On that basis, a two-cluster solution was selected for the final K-means analysis.

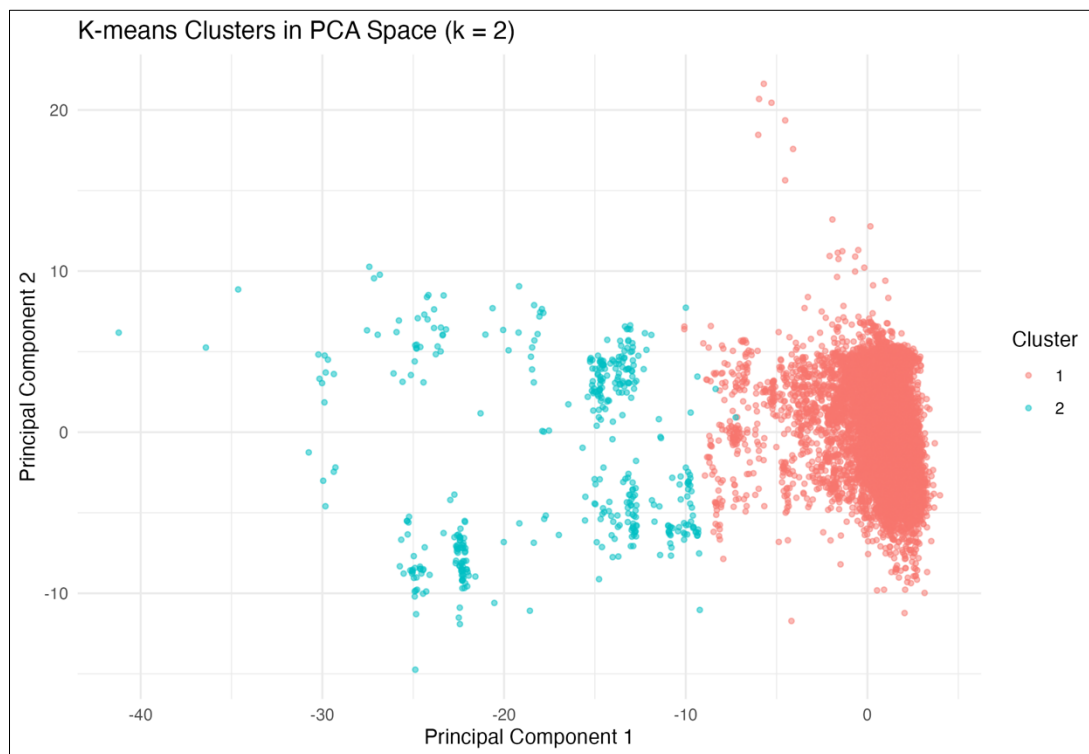


Figure 3 - K-means clusters in PCA space

A two-cluster specification was then fitted to a reproducible sample of the scaled data. When visualised in PCA space, the resulting clusters show a reasonably coherent division of the sampled observations into two broad similarity groups.

These clusters should not be interpreted as direct categories of criminality. Rather, they represent similarity groupings within the broader feature space. Their value is exploratory: they show that the data contain non-trivial structure, and they provide a first indication that machine learning methods may be able to extract meaningful patterns from the transaction attributes.

4. Regression Analysis

For the regression task, a continuous target variable was required. Because the dataset does not include a natural continuous risk outcome, a proxy transaction risk score was constructed. Illicit transactions were assigned a score of 1.0, licit transactions a score of 0.0, and unknown transactions a score of 0.5. This score should not be read as a real observed measure of financial risk. It is an analytical device intended to represent relative review priority and to create a meaningful basis for model comparison.

Four regression models were tested: linear regression, ridge regression, lasso regression, and random forest regression, combining standard linear benchmarks with a more flexible nonlinear ensemble method (James et al., 2014; Breiman, 2001). The first three models provide useful baselines, while the random forest model allows more flexible nonlinear relationships between the transaction features and the proxy score to be captured than would be possible under a purely linear specification (James et al., 2014; Breiman, 2001).

Table 3 - Regression results

Model	RMSE	MAE	R2
Linear Regression	0.1842	0.112	0.3046
Ridge Regression	0.1808	0.1134	0.3268
Lasso Regression	0.1799	0.1119	0.3334
Random Forest Regression	0.0984	0.0432	0.8036

The results show a clear ordering. Linear regression performs worst, while ridge and lasso improve modestly on the linear baseline. Random Forest Regression performs substantially better than the other three models, achieving the lowest RMSE and MAE and the highest R^2 . This suggests that the relationship between the transaction features and the constructed score is not well captured by a simple linear form, and that nonlinear structure or interaction effects may be important in the data (Breiman, 2001; James et al., 2014).

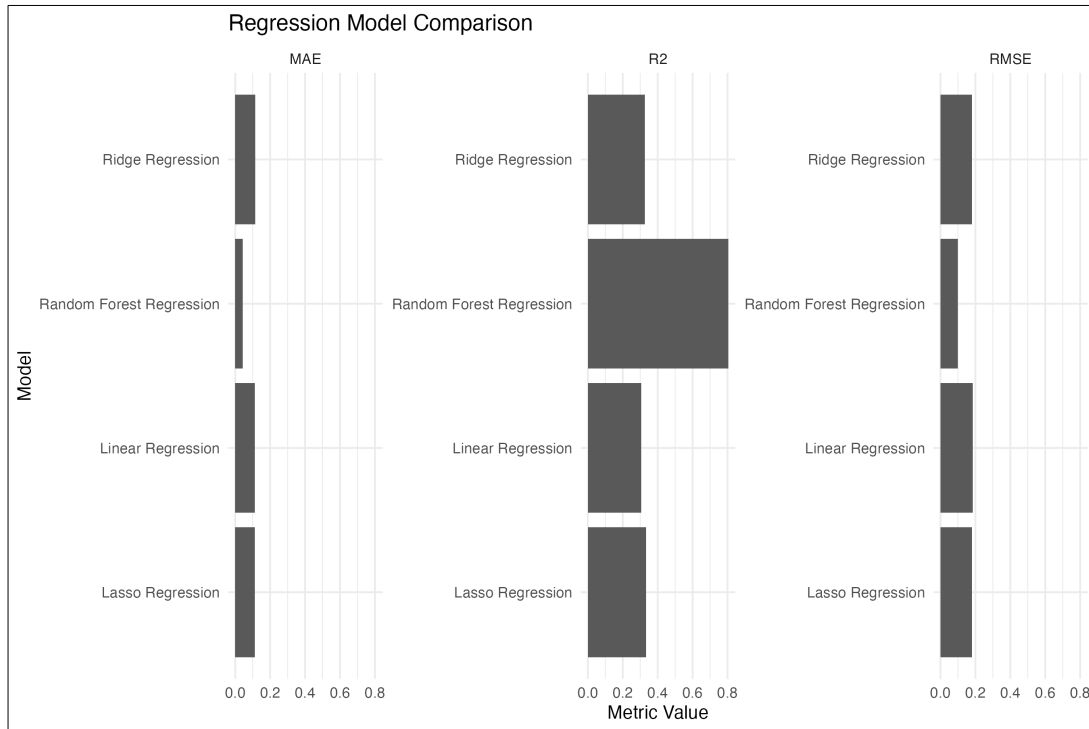


Figure 4 - Regression model comparison

The regression results should be interpreted carefully. Because the target variable is engineered rather than observed, the aim is not causal explanation or economic forecasting, but rather a comparative modelling exercise based on an analytically constructed notion of transaction priority (Murphy, 2012; James et al., 2014). Instead, the regression exercise shows whether different models can reproduce an internally defined notion of relative transaction priority. In that more limited but still useful sense, the random forest model provides the strongest analytical fit.

5. Classification Analysis

The classification task focuses on the labelled subset of the dataset and asks which model best distinguishes illicit from licit transactions. This is the most directly operational part of the project, since the practical question is whether potentially illicit transactions can be identified with useful accuracy.

Three classification models were compared: logistic regression, Support Vector Machine, and Random Forest, representing three widely used approaches to supervised classification in applied machine learning (Cortes and Vapnik, 1995; Breiman, 2001; James et al., 2014). Because the labelled data are imbalanced, model performance was assessed not only through accuracy, but also through precision, recall, and F1-score, with particular attention to recall for the illicit class, since minority-class detection is more informative here than overall accuracy alone (James et al., 2014). In an AML context, a model that misses most illicit cases would have limited usefulness even if its overall accuracy appeared high.

Table 4- Classification results

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.9036	0.7391	0.0187	0.0365
Support Vector Machine	0.9712	0.9245	0.7679	0.8389
Random Forest	0.9868	0.9937	0.8702	0.9279

The comparison reveals an important contrast between overall accuracy and class-specific performance. Logistic regression achieves apparently respectable accuracy but performs very poorly in identifying illicit transactions. This is a classic example of the distortion created by class imbalance, where a model can appear accurate overall while still performing poorly on the class of greatest practical interest (James et al., 2014). Support Vector Machine performs much better, with a substantial improvement in both recall and precision for the illicit class. Random Forest Classification performs best overall, producing the highest accuracy, strongest F1-score, and the highest recall for illicit transactions among the models tested, which is consistent with the strong predictive performance often achieved by ensemble tree methods on structured tabular data (Breiman, 2001).

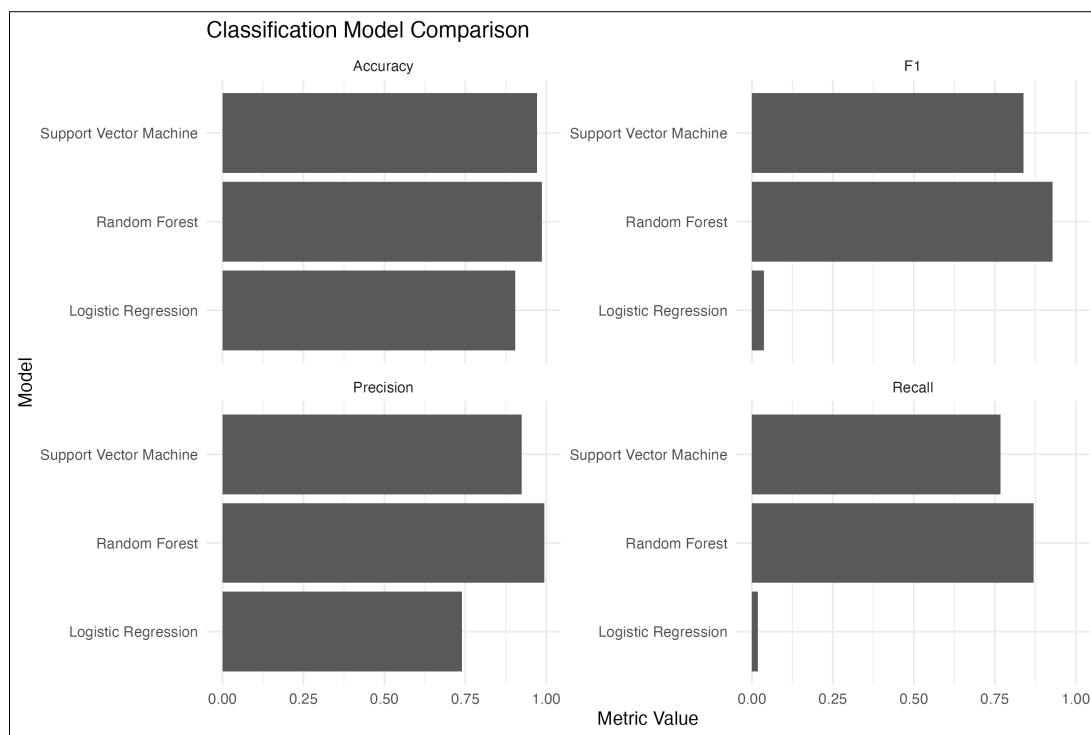


Figure 5 - Classification model comparison

The Random Forest model was therefore selected as the best classifier. Its confusion matrix shows that it correctly identifies 791 illicit transactions and 8398 licit transactions, while missing 118 illicit cases and wrongly flagging only 5 licit cases as illicit. This is a strong result because it combines high illicit detection with a very low false-positive rate.

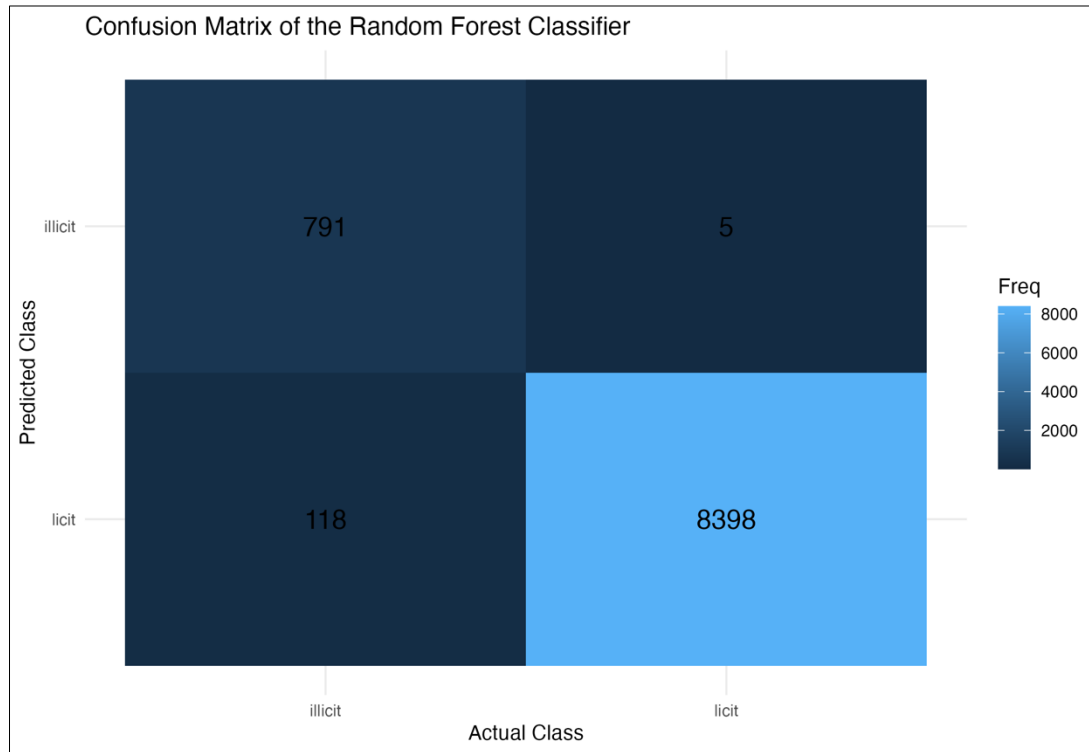


Figure 6 - Confusion matrix of the Random Forest classifier

To add interpretive depth, feature importance was examined using Mean Decrease in Gini, a common variable-importance measure derived from Random Forest models (Breiman, 2001). The results indicate that predictive performance is concentrated in a relatively small subset of variables rather than being spread evenly across all transaction features.

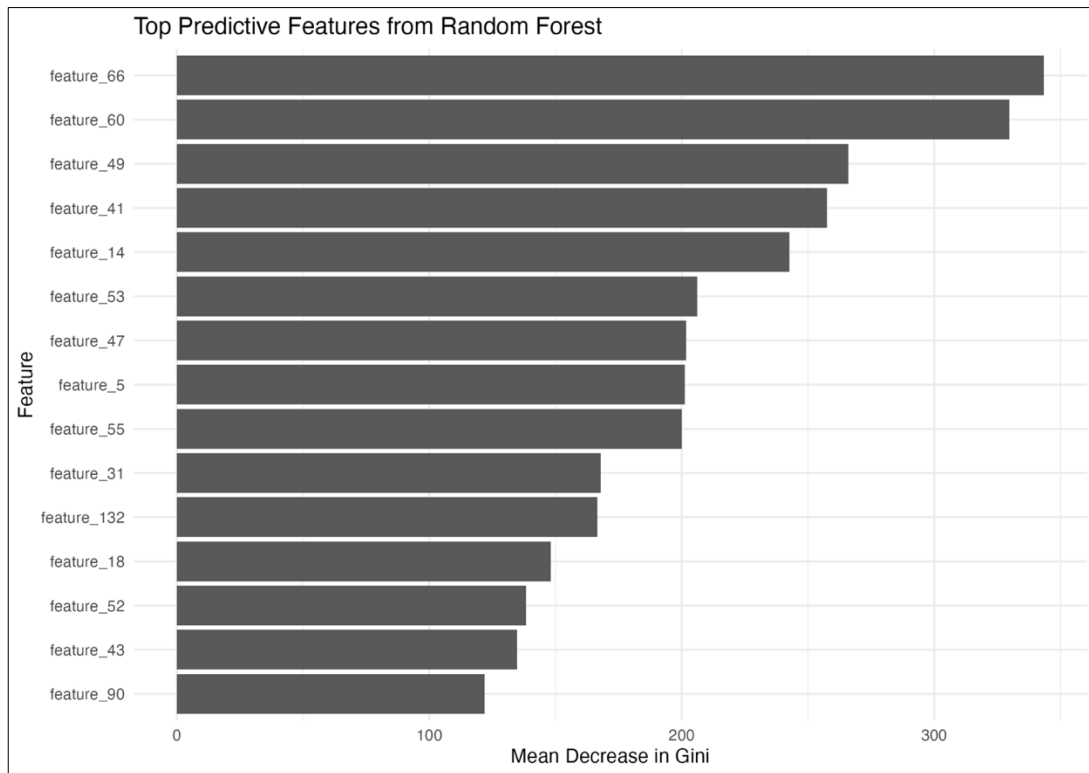


Figure 7 - Top predictive features from the Random Forest classifier

The most important variables include feature_66, feature_60, feature_49, feature_41, and feature_14. These results should not be interpreted causally, but they do suggest that a limited number of transaction characteristics carry much of the predictive signal used by the classifier, rather than predictive value being spread evenly across all available features (Breiman, 2001).

Table 5- Top ten predictive features from the Random Forest classifier

Feature	feature_66	feature_60	feature_49	feature_41	feature_14	feature_53	feature_47	feature_5	feature_55	feature_31
Importance	343.3739	329.6845	265.9726	257.4762	242.5848	206.1726	201.6752	201.1986	200.1089	167.9692

6. Discussion

Taken together, the three analytical tasks provide a coherent picture of the Elliptic dataset. The unsupervised analysis shows that the transaction feature space contains meaningful structure, although that structure does not translate directly into a simple visual separation between illicit and licit activity. The regression analysis suggests that nonlinear methods are better suited than standard linear approaches when the aim is to approximate a transaction-priority score from the available features. The classification analysis provides the strongest evidence of practical value, especially through the performance of the Random Forest model on the illicit minority class.

At the same time, several limitations should be acknowledged. First, the regression target is constructed rather than observed, so the regression section should be interpreted as an exercise in analytical prioritisation rather than real-world prediction. Second, the transaction features are engineered variables, which means the analysis works with a transformed representation of behaviour rather than raw blockchain activity. Third, even a strong classifier should not be treated as proof of wrongdoing. In practice, such a model would be more appropriately understood as a screening or prioritisation tool to support further review, rather than as a substitute for human judgment in financial-crime analysis (Baesens, Van Vlasselaer and Verbeke, 2015; Weber et al., 2019).

These limitations do not weaken the overall value of the project; rather, they clarify where machine learning is most useful in an anti-money-laundering context: as a tool for prioritisation, triage, and analytical support in large transaction environments, rather than as a substitute for professional judgment.

7. Conclusion

This project applied unsupervised learning, regression, and classification methods to the Elliptic Bitcoin Dataset in order to examine the detection and prioritisation of potentially illicit cryptocurrency transactions. The unsupervised analysis showed that the data contain meaningful structure, while also indicating that similarity-based groupings should not be interpreted as direct categories of criminality. In the regression task, Random Forest Regression clearly outperformed linear, ridge, and lasso models when predicting the proxy risk score. In the classification task, Random Forest Classification delivered the strongest results, combining high recall for illicit transactions with very low false-positive rates.

Overall, the findings suggest that machine learning can provide useful analytical support for identifying and prioritising suspicious cryptocurrency transactions, particularly when used as part of a broader analytical process rather than as a stand-alone decision mechanism (Baesens, Van Vlasselaer and Verbeke, 2015; Weber et al., 2019). At the same time, the project also shows that model results must be interpreted with care, especially where labels are incomplete and the operational meaning of prediction is sensitive, as is often the case in real-world anti-money-laundering and financial-intelligence settings (Weber et al., 2019). Within those limits, the analysis provides a defensible example of how machine learning methods can contribute to financial intelligence and anti-money-laundering work.

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